

(a) Augmented variants from one training image
source: train/brown_spot/100001.jpg

(b) Encoded JSON output
produced by the system's forecast service

augmentation applied at the 224 × 224 px model input resolution used in training; panels enlarged for display

Original



no augmentation

Rotation



rotation +25° (range ±25°)

Horizontal flip



horizontal flip (seed 0)

Vertical flip



vertical flip (seed 0)

Random crop



random crop, scale 0.8
(range 0.8–1.0)

Gaussian noise



Gaussian noise, variance 0.01 on the [0, 1] pixel scale

Full pipeline #1



full pipeline, random draw (seed 42)

Full pipeline #2



full pipeline, random draw (seed 43)

```
{
  "confidence": {
    "Healthy": 3.71008318325039e-05,
    "High Stress": 0.9997690320014954,
    "Moderate Stress": 0.00019386109488550574
  },
  "date": "2024-11-02 08:54:53",
  "forecast_type": "forecast",
  "humidity": 47.24784315403373,
  "light_intensity": 760.6258292666594,
  "nitrogen": 31.052469900815304,
  "ph": 7.497374386873277,
  "phosphorus": 30.663180860058453,
  "potassium": 33.08383762021031,
  "predicted_health": "High Stress",
  "soil_moisture": 36.35755797515368,
  "soil_temp": 20.280489828192664,
  "temperature": 23.021988353360015
}
```